# Using u-Substitution in Indefinite and Definite Integrals
# Answer Key

AP Calculus AB / BC Integration Techniques

### Quick Answers

1. $\frac{\left(x^2+5\right)^4}{4}$
2. $\frac{\sin\left(x^2\right)}{2}$
3. $\frac{\log(x)^3}{3}$
4. $-\frac{1}{2}+\frac{e^4}{2}$
5. $\frac{\sin^4(x)}{4}$
6. $\log(5)$
7. $\frac{2\left(x^3+1\right)^{\frac{3}{2}}}{9}$
8. $\frac{1}{2}$, $\frac{1}{2}$

### Curriculum

**Level:** AP Calculus AB / BC Integration Techniques
FUN-6.D–FUN-6.E – *problems 1, 2, 3, 4, 5, 6, 7, 8*
*Difficulty 1–5 of 5 across 9 tagged checks.*

**1.** $\displaystyle\int 2x\left(x^2+5\right)^3 dx$.

**Step 1.** Look for an inner function whose derivative also appears. The inner function is $x^2+5$, so let $u = x^2+5$; then $du = 2x\,dx$, which is exactly the factor sitting outside the bracket.

**Step 2.** The integral becomes $\displaystyle\int u^3\,du = \frac{u^4}{4}$, and back-substituting returns the answer in $x$.

$$\boxed{\frac{(x^2+5)^4}{4}+C}$$

*Check by differentiating:* $\frac{4(x^2+5)^3\cdot 2x}{4} = 2x(x^2+5)^3$, the original integrand.

**2.** $\displaystyle\int x\cos\left(x^2\right) dx$.

**Step 1.** Let $u = x^2$, so $du = 2x\,dx$. The integrand offers only $x\,dx$, which is $\frac{1}{2}\,du$ — a constant short, not a function short.

**Step 2.** Part (b): a constant may be moved across the integral sign because $\int kf = k\int f$; a variable may not, since $x$ changes as the integration variable moves. So $\displaystyle\int x\cos(x^2)\,dx = \frac{1}{2}\int \cos u\,du = \frac{1}{2}\sin u$.

$$\boxed{\frac{\sin(x^2)}{2}+C}$$

*Check:* $\frac{1}{2}\cos(x^2)\cdot 2x = x\cos(x^2)$.

**3.** $\displaystyle\int \frac{(\ln x)^2}{x}\,dx$.

**Step 1.** Part (a): $\ln x$ is the inner function and $\frac{1}{x}$ is its derivative, so the integrand is already in the form $f(u)\,du$.

**Step 2.** With $u = \ln x$ and $du = \frac{1}{x}\,dx$, the integral is $\displaystyle\int u^2\,du = \frac{u^3}{3}$.

$$\boxed{\frac{(\ln x)^3}{3} + C}$$

*Check:* $\frac{3(\ln x)^2}{3} \cdot \frac{1}{x} = \frac{(\ln x)^2}{x}$.

**4.** $\displaystyle\int_0^2 x\,e^{x^2}\,dx$.

**Step 1.** Let $u = x^2$, so $du = 2x\,dx$ and $x\,dx = \frac{1}{2}du$. Because the integral is definite, convert the limits too: $x = 0$ gives $u = 0$, and $x = 2$ gives $u = 4$.

**Step 2.** The integral becomes $\displaystyle\frac{1}{2}\int_0^4 e^u\,du = \frac{1}{2}\left[e^u\right]_0^4 = \frac{e^4 - 1}{2}$.

$$\boxed{\frac{e^4 - 1}{2}}$$

*Why convert the limits:* once the bounds are in $u$, no back-substitution is needed. Evaluating $\left[\frac{1}{2}e^u\right]$ at the *old* bounds 0 and 2 is the single most common error on this problem type.

**5.** $\displaystyle\int \sin^3 x\,\cos x\,dx$.

**Step 1.** Part (a): with $u = \sin x$, $du = \cos x\,dx$; with $u = \cos x$, $du = -\sin x\,dx$. Test each against what the integrand actually supplies before committing.

**Step 2.** Part (b): $u = \sin x$ consumes the whole integrand, since $\sin^3 x\,\cos x\,dx = u^3\,du$. The other choice absorbs only one $\sin x$ and strands $\sin^2 x \cos x$ behind, which cannot be written in $u$ alone.

**Step 3.** $\displaystyle\int u^3\,du = \frac{u^4}{4}$, then back-substitute.

$$\boxed{\frac{\sin^4 x}{4} + C}$$

*Check:* $\frac{4\sin^3 x \cdot \cos x}{4} = \sin^3 x \cos x$.

**6.** $\displaystyle\int_1^3 \frac{2x}{x^2+1}\,dx$.

**Step 1.** Let $u = x^2 + 1$, so $du = 2x\,dx$ — the numerator exactly. New limits: $x = 1$ gives $u = 2$, and $x = 3$ gives $u = 10$.

**Step 2.** $\displaystyle\int_2^{10} \frac{du}{u} = \left[\ln|u|\right]_2^{10} = \ln 10 - \ln 2$.

**Step 3.** Part (c): a difference of logarithms is the logarithm of the quotient, and $10 \div 2 = 5$.

$$\boxed{\ln 5}$$

*Note on absolute values:* on the new interval $u$ runs from 2 to 10, so $u > 0$ throughout and the absolute value can be dropped without comment.

**7.** $\displaystyle\int x^2\sqrt{x^3+1}\,dx$**.**

**Step 1.** Let $u = x^3+1$, so $du = 3x^2\,dx$ and $x^2\,dx = \frac{1}{3}du$. The integral carries a factor of one third.

**Step 2.** Part (b): write the root as a power — $\displaystyle\frac{1}{3}\int u^{1/2}\,du$ — so the power rule applies.

**Step 3.** $\frac{1}{3}\cdot\frac{u^{3/2}}{3/2} = \frac{1}{3}\cdot\frac{2}{3}u^{3/2} = \frac{2}{9}u^{3/2}$. $\boxed{\dfrac{2\left(x^3+1\right)^{3/2}}{9}+C}$

*Check:* $\frac{2}{9}\cdot\frac{3}{2}(x^3+1)^{1/2}\cdot 3x^2 = x^2\sqrt{x^3+1}$.

**8. Challenge:** $\displaystyle\int_0^\infty x\,e^{-x^2}\,dx$**.**

**Step 1.** Part (a): infinity is not a value the Fundamental Theorem can be evaluated at, so the integral is *defined* as $\displaystyle\lim_{b\to\infty}\int_0^b x\,e^{-x^2}\,dx$. Everything else is an ordinary substitution.

**Step 2.** Let $u = -x^2$, so $du = -2x\,dx$ and $x\,dx = -\frac{1}{2}du$. The antiderivative is $-\frac{1}{2}e^{-x^2}$, and on $[0,b]$ it evaluates to $-\frac{1}{2}e^{-b^2}+\frac{1}{2}$.

**Step 3.** Part (b): as $b\to\infty$, $e^{-b^2}\to 0$, so the limit is the constant term that survives. $\boxed{\dfrac{1}{2}}$

**Step 4.** Part (c): the limit exists and is finite, so the improper integral converges, and its value is that limit. $\boxed{\dfrac{1}{2}}$

*Why the substitution still governs:* the limit only decides convergence. The number itself came from the antiderivative $u$-substitution produced — and note that $u = x^2$ works just as well, giving $\frac{1}{2}\left(1-e^{-b^2}\right)$, the same expression rearranged.